

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior full-stack TypeScript / Node.js engineer with 6+ years shipping data-heavy APIs and user-facing features. Led the zero-downtime migration of 20+ production apps at MTailor, designing a Pub/Sub-based sync pipeline across MongoDB and Firestore and cutting infrastructure costs significantly. Built a WhatsApp healthcare chatbot integrating OpenAI and Google Vision for prescription OCR. At Dr Wealth, queried 2M+ Firestore records per run to keep a real-time PWA current. Comfort bridging backend inference services and React UIs — the full stack of AI/ML features.

CORE COMPETENCIES

TypeScript / Node.js

Full-stack AI/ML features

LLM integration

Scalable APIs

React

Query optimization

Real-time analytics

Cloud / distributed systems

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote

- Led end-to-end migration of 20+ applications from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to the CTO.
- Built a real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub — idempotent dual-write over live production traffic.
- Shipped a buyer-facing 3D visualization feature with ffmpeg video overlay — increased conversion.
- Migrated file storage from S3 to GCS via Python script; cut \$5,000/month by moving off AWS.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Built ad-hoc jobs querying 2M+ Firestore records — query optimization for a real-time, user-facing PWA.
- Extended backend APIs with Firebase Cloud Functions + Express.js on Heroku.
- Shipped responsive UI pages in React + Tailwind.

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer — US Remote (Agency)

- Built backends for DeFi applications with Web3.js + Node.js.
- Delivered React UIs from Figma under tight agency deadlines.

CodeBits

Jan 2020 – Jul 2021

Software Engineer — Uganda Remote (Led team of 4)

- Architected a Node.js microservices backend (Kafka + gRPC + NATS Streaming + Docker + Kubernetes) for FIDA Uganda's case-management app.
- Built React Native + Expo mobile app for LASPNET with Firebase push notifications.
- Built a USSD service bridging to the backend over gRPC.

MTailor (Contractor)

Mar 2022 – Jul 2022

Contractor — US Remote

- Built migration script reading MongoDB and writing to Firestore after complex processing; authored SDK docs consumed by fellow engineers and Ops.

PROJECTS

Healthcare WhatsApp Chatbot (OpenAI + Google Vision)

End-to-end LLM integration — prescription OCR over WhatsApp Business API, with OpenAI for natural-language flow and Google Vision for image parsing.

Node.js · OpenAI · Google Vision · WhatsApp Business API

Real-Time Pub/Sub Sync Pipeline (MTailor)

Idempotent MongoDB ↔ Firestore sync supporting parallel production traffic during migration of 20+ apps.

Node.js · Google Pub/Sub · Firestore · MongoDB

Marketplace Recommendation Engine

Collaborative + content-based filtering engine in MongoDB for service-provider matching.

Node.js · MongoDB · TypeScript

EDUCATION

B.Sc. Computer Engineering · [Makerere University](#)

Aug 2018 – Sep 2022

SKILLS

Proficient: Node.js, TypeScript, React, Firebase, GCP, Python, MongoDB, SQL, Git, Linux

Intermediate: Go, AWS, Rust, React Native, Kubernetes, Kafka, OpenAI API

Languages: English · Kiswahili